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Glycyrrhiza uralensis Fisch. alleviates dextran sulfate sodium-induced colitis in mice through inhibiting of NF-κB signaling pathways and modulating intestinal microbiota

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ABSTRACT

Ethnopharmacological relevance: Licorice is widely used in traditional Chinese Medicine (TCM) for compound compatibility, which could reduce toxicity and increase efficacy of certain herbal medicine, and its active components prominently effects of inhibit of inflammation and regulate of immunity.

Aim of the study: The study probed into the mechanism of the anti-inflammatory and immunomodulatory effects of licorice based on the domination of the T helper type 17/regulatory T cells (Th17/Treg) differentiation balance and the composition and structure of the intestinal flora through the nuclear factor kappa B (NF-κB) signaling pathway.

Materials and methods: BALB/c mice were inoculated with dextran sulfate sodium (DSS) to establish animal models of ulcerative colitis (UC). For the pharmacodynamic study, UC mice were observed for the anti-inflammatory effect of licorice water extraction (LWE) *in vivo*, including clinical observation and measurement of colon length. Hematoxylin-eosin (HE) staining was used to evaluate pathological conditions. Immunohistochemistry (IHC) and transmission electron microscopy (TEM) were performed to observe the intestinal barrier of the colons. Inflammatory cytokine levels were measured using with enzyme-linked immunosorbent assay (ELISA) kits. The proportions of T helper (Th) cells in the colons was assessed using flow cytometry. Gut microbiota diversity was detected using 16S ribosomal (r)DNA sequencing. In addition, Western blot (WB) assays were used to verify ROR-γt, Foxp3, TLR4, MyD88 and NF-κB expression according to a standard protocol.

Results: LWE exerted a pharmacological anti-inflammatory effect by attenuating inflammation in the colonic tissues through affecting the protein expression of TLR4/MyD88/NF-κB, and increasing the expression of tight junction (TJ) protein in the colons, improving the integrity of the intestinal mucosal barrier *in vivo*. Moreover, LWE reversed the imbalance in Th17/Treg cells differentiation and influenced the protein expression of ROR-γt and Foxp3 in UC mouse colons. In particular, LWE significantly affected the diversity of the gut microbiota in UC mice, ameliorated the composition of dominant species, and significantly increased the type and quantity of probiotics.

Conclusion: Licorice tends to reduce inflammation and enhance the protective action of the intestinal mucosal barrier via the TLR4/MyD88/NF-κB signal transduction pathway and alter the imbalance of Th-cell differentiation. Notably, licorice may affect the diversity of intestinal microbiota and the content of beneficial bacteria in the colon, which is a potential mechanism for understanding anti-inflammatory and immunomodulatory effects in UC mice *in vivo*.

1. . Introduction

In recent years, ulcerative colitis (UC), which is an inflammatory bowel disease (IBD), has been characterized by chronic relapsing colonic inflammation, disorganized immune responses in the intestinal tract, and various complications, posing a crucial health problem that causes substantial morbidity worldwide (Du and Ha, 2020; Kaenkumchorn and

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Abbreviations

IBD	Inflammatory Bowel Diseases
CD	Crohn's Disease
UC	Ulcerative Colitis
DSS	Dextran Sulfate Sodium
TCM	Traditional Chinese Medicine
LWE	Licorice Water Extraction
HPLC	High Performance Liquid Chromatography
ELISA	Enzyme-Linked Immunosorbent Assay
IL-1 β	Interleukin-1 beta
IL-8	Interleukin-8
IL-18	Interleukin-18
TNF-α	Tumor Necrosis Factor
IL-10	Interleukin-10
Th17	T helper type 17
Treg	regulatory T cells
IHC	Immunohistochemistry
PBS	phosphate-buffered saline
TEM	Transmission Electron Microscopy
HE	Hematoxylin-eosin
TJ	tight junction
ZO-1	Zonula occludens-1
SDS	Sodium Dodecyl Sulfate
OTUs	Operational Taxonomic Units
WB	Western blot
PCoA	Principal Coordinates analysis
PLS-DA	Partial Least Squares Discriminant Analysis
TLR4	Toll-like receptor 4
MyD88	Myeloid Differentiation primary response gene 88
NF-κB	Nuclear Factor kappa B
ANOVA	One-way Analysis of Variance
COG	Clusters of Orthologous Groups
KEGG	Kyoto Encyclopedia of Genes and Genomes

Wahbeh, 2020; Yu and Rodriguez, 2017). Evidence has suggested that the etiology of UC involves environmental, genetic, immunological and infectious factors. It is essential to pay attention to the dysregulation of the immune system, especially the differentiation and activation of immune cells leading to intestinal inflammation, as intestinal microbial dysbiosis could exacerbate the development of the disease (Glassner et al., 2020; Hanssen et al., 2021; Kobayashi et al., 2020).

Accumulating evidence suggests that UC arises from a complex interaction among inflammation, which is one of the most common symptoms in UC, immune responses, and microbial flora factors. Inflammatory responses and immune imbalances can be adjusted by modifying the concentrations of proinflammatory cytokines and signaling pathways (Chen et al., 2020a; Leppkes and Neurath, 2020). One study assessed the mechanism underlying the development of UC and UC-associated cancers, revealing the importance of inflammation in the process of UC (Yao et al., 2019). Recent experimental investigations have indicated that the intestinal microflora plays a critical role in the regulation of host immunoinflammatory reactions and is closely associated with intestinal diseases. Accordingly, dysbiosis of the intestinal microorganisms has been observed in multiple inflammatory and/or immune-induced diseases involving UC (Aldars-Garcia et al., 2021; Tan et al., 2020).

Licorice, the roots and rhizomes of *Glycyrrhiza uralensis* Fisch., *Glycyrrhiza glabra* L. and *Glycyrrhiza inflata* Batalin. is a well-known traditional perennial herbaceous shrubs, it was originally recorded in Shen Nong's herbal classic (Shen Nong Ben Cao Jing) during the Han Dynasty. According to the basic theory of traditional Chinese medicine (TCM), licorice was traditionally reported to be used for tonifying middle-Jiao and Qi (He-Zhong-Yi-Qi) by recorded in Mingyi Bielu and Collected Statements on the Herbal Foundation (Bencao Huiyan), which the "middle-Jiao" refers to the digestive system. In addition, the Chinese Pharmacopoeia 2020 edition also mentioned that licorice has the effects of tonifying effects on the spleen and replenishes qi (Bu-Pi-Yi-Qi), clears away heat and detoxification, relieves pain, and reduces sore carbuncle swelling toxin. It has been used for thousands of years to treat a variety of inflammatory diseases and shows comparable efficacy of immunor egulation, and it enhances the effectiveness of TCM as a special "guide drug", such as Banxia Xiexin Decoction (Zhang, 1997), Baitouweng Decoction (Zhang, 1983), and Sini Powder (Chen, 1996). It is worth noting that since the 1950s, researchers in some regions of the world have reported that licorice extract has a significant therapeutic effect on ulcerative diseases (Akagi et al., 1971; Ishii and Fujii, 1982; Ye et al., 1958). In clinical practice, medical researchers have found that compound glycyrrhizin injection has a positive effect on the treatment of UC (Chang, 2004; Xiang, 2006). Licorice constituents are recommended in the European Union for the treatment of gastrointestinal disorders based on traditional use (Thumann et al., 2019). Additionally, scholars traced the evolution of herbal medicine (include licorice) in Russia, elaborately summarized and critically assessed the 14th edition of the State Pharmacopoeia of the Russian Federation, pointed out that all medicines of licorice are available in Russia in Pharmacies as OTC products and it is often used as a food additive locally (Shikov et al., 2017, 2021). From the 21st century, with the continuous renewal of research methods for medicinal botany, the major active components extracted from liquorice primarily include glycyrrhizin, liquiritin, and glycyrrhetinic acid (Heidari et al., 2021). The hydrolyzed metabolite of glycyrrhizin, 18α-Glycyrrhetinic acid owns effect of anti-inflammatory activities (Richard, 2021; Wu et al., 2022). Glycyrrhizin exhibits remarkable biological activity in the human body by affecting the gut flora and intestinal barrier (Qiu et al., 2019; Yu et al., 2018).

A preliminary experimental study of the research group found that licorice water extraction (LWE) showed a significant protective effect in UC mice. Nevertheless, the mechanism of action of LWE remains unclear. To further explore the effect of licorice in UC mice, this study focused on the survey of flora diversity and Th cells to elucidate the positive role of LWE on the intestinal microecology of UC mice and to provide an experimental basis for uncovering its possible mechanism of action.

2. .Materials and methods

2.1. Chinese herbal medicine material and chemical detection reagent

The roots of licorice were purchased from Anhui Guanghe Chinese Medicine Co., Ltd. (Bozhou, China), and the plant name has been confirmed through "The Plant List" (www.theplantlist.org). Prof. Chengwu Fang from Anhui University of Chinese Medicine identified licorice according to the Chinese Pharmacopoeia 2020 edition. Dextran sulfate sodium (DSS) was purchased from Yisheng Biotechnology Company (Shanghai, China). Mouse IL-1β, IL-8, IL-18 and TNF-α ELISA kits were purchased from Jingmei Biotechnology Co., Ltd. (Jiangsu, China). Sustained release mesalazine granules was obtained from Aidifa Pharmaceutical Co., Ltd. (Shanghai, China). Mouse TLR4 and MyD88 antibodies were purchased from Affinity Biosciences (Jiangsu, China), Occludin, ZO-1 and nuclear factor kappa B (NF-κB) p65 antibodies were obtained from Abcam (Cambridge, MA, USA), Claudin-1 antibody was purchased from Abclonal (Wuhan, China), and ROR-γt and Foxp3 antibodies were purchased from ZenBio Antibody (Chengdu, China).

2.2. Extraction of licorice and major component analysis

The licorice was extracted using the cold percolation method. Accurately weigh 80 g licorice, add 10 times the volume of distilled water. The solvent was evaporated in a boiling water bath, and the extract was concentrated in a vacuum rotary evaporator below 45 °C, extracted twice and concentrated to 50 mL to obtain 1.6 g/mL stock solution, stored at −20 °C.

High performance liquid chromatography (HPLC) was performed after filtration of LWE using a 0.22 μm microporous membrane. An Agilent WondaSil C18 Superb analyzer was used for the chromatographic separations. The mobile phase was composed of acetonitrile (A) and 0.1% phosphoric acid water with a linear gradient elution as follows: 0–15 min, 20%–30% A; 15–25 min, 30%–32% A; 25–30 min, 32%–43% A; 30–43 min, 43%–44% A; 43–48 min, 44%–90% A; and 48–55 min, 90%–20% A. The eluate was monitored using a diode array detector at a wavelength of 254 nm (20 μL, 1.0 mL/min, and 30 °C).

2.3. Animal grouping and DSS-induced UC

Sixty healthy 6–8 weeks old male BALB/C mice (20 ± 2 g) were purchased from Jiangsu Huangchuang Sino Pharmaceutical Technology Co., LTD (SCXK (Su) 2020–0009, Jiangsu, China). The mice were subjected to a 12 h light/dark cycle in a specific pathogen-free facility, and sterilized food and water were obtained from the laboratory animal center of the university. This research was approved by the university's Institutional Animal Ethics Committee of the university, and experiments were conducted according to the ethical principles of experimental animals (Approval No. 2022007).

The mice were reared adaptively for 7 d and randomly separated into the control group (normal mice), model group (3% DSS), mesalazine group (3% DSS + mesalazine), LWE high-dose group (8 g/kg/d), LWE medium-dose group (4 g/kg/d), and LWE low-dose group (2 g/kg/d), each group with ten mice. Excluding the control group, mice in all other groups were free to drink 3% DSS solution (freshly prepared daily; 3 g DSS was accurately weighed and dissolved in 100 mL of distilled water). During this period, the mice were free to eat and drink. Treatment started on the 8th day. The LWE groups were administered licorice water extract (8, 4, and 2 g/kg/d) by gavage for 7 d, the mesalazine group were administered 200 mg/kg/d mesalazine, and the other mice were treated with a suitable volume of distilled water.

2.4. Clinical observation and measurement of colon length

The physiological status of each group of mice, including mental state and activities, food and water intake and body weight, was regularly observed and recorded daily. Each recording was performed before and after drug administration. After 7 d of drug intervention, all mice were anesthetized with 2% pentobarbital sodium (30 mg/kg), and blood samples were collected by extracting the eyeball before fasting for 12 h. Mice were sacrificed through cervical dislocation, and a laparotomy was promptly performed to clear and measure the colon length.

2.5. Pathological evaluation and immunohistochemistry (IHC) of colons

Samples were fixed overnight in 4% paraformaldehyde for histopathological assessment and IHC. The colons tissue was removed, and the contents were cleaned using an ice-cold saline enema. In each group, 1 cm of colon tissue was cut from the same site, fixed in 10% formalin fixative and gradient concentration ethanol solution, and then permeabilized to transparency in xylene. The sections were stained with hematoxylin dye and then dried for observation and photography using a microscope.

Colonic tissues were embedded in paraffin, and immunohistochemical localization was performed after ultrathin section. IHC was performed after the tissue sections were incubated with 3% H_2O_2 for 20 min in order to remove endogenous tissue peroxidase, after washing with phosphate-buffered saline (PBS) thrice and incubating them with the primary antibodies (occluding [1:100], claudin-1 [1:500], ZO-1 [1:500]) at 37 °C for 60 min. After the secondary antibody had been added, the sections were incubated at 37 °C in a humidified chamber for 20 min and analyzed using a microscope after being development with DAB chromogen at 22 °C.

2.6. Transmission electron microscopy (TEM) observation of the intestinal barrier

For TEM observation, 5 mm^2 pieces were cut from each mouse colon and fixed in 2.5% glutaraldehyde (prepared with sterile cold PBS) at 4 °Covernight. Then, fixation with 1% osmium tetroxide for 1 h, and step-by-step dehydration with gradient diluted ethanol (30%, 50%, 70%, 80%, 95%, 100%, each concentration for 15 min, repeated thrice) were performed. The samples were soaked for 1 h in 50% epoxy propylene and then transferred to Epon812 for overnight embedding. Ultrathin sections (70 nm thickness) were obtained using an ultramicrotome (UC-7, Leica, Germany). After staining with uranyl acetate and lead acetate, the intestinal villi and crypt structure were photographed using TEM (JEM-1400, JEOL, Japan).

2.7. Cytokine determination

Blood was extracted using centrifugation in a cryogenic centrifuge for 15 min after homogenization, and the serum was stored at in an ultra cold storage freezer. The concentration of cytokines was examined for the IL-1β, IL-8, IL-18, TNF-α and IL-10 was analyzed using the mouse ELISA kit, according to the manufacturer's instructions.

2.8. Flow cytometry

Colonic tissues were separated using a tissue lysis kit in a dissociator, and cell suspension of colonic tissue was obtained. After cell incubation, the cells were collected and centrifuged at 1000 rpm for 10 min and resuspended in PBS, and incubated with antibodies (anti-CD4-FITC or anti-CD25-FITC). After that, the cells were fixed with 2% formaldehyde and infiltrated with 0.1% Triton X-100. Thereafter, anti-IL-17 or anti-Foxp3 antibodies were added and the cells were incubated for 1 h. The T helper cells were selected by flow cytometry, and the ratio of Th17 to regulatory T (Treg) cells was calculated after setting the cross gate.

2.9. Western blot (WB) analysis

After the colon tissue was cut into pieces, 1 mL of NP-40 lysis buffer (containing 1 mM PMSF) was added for homogenization, and centrifuged at 12,000 rpm for 30 min to collect the supernatant. Ten micrograms of protein from each sample were loaded onto sodium dodecyl sulfate (SDS) polyacrylamide gels and transferred to a nitrocellulose imprinted membrane. These membranes were then blocked with freshly prepared 5% nonfat dry milk at low temperature and incubated with antibodies (anti-Toll-like receptor 4 [TLR4), anti-myeloid differentiation primary response gene 88 [MyD88], anti-nuclear factor kappa-B [NF-κB] p65, anti-forkhead box protein 3 [Foxp3], or anti-retinoid-related orphan nuclear receptor γt [ROR-γt]) overnight. Finally, horseradish peroxidase was added and the membranes was incubated for 1 h, and the immune response band was detected using an ECL detection kit. Images were analyzed quantitatively using the Image J software.

2.10. Gut microbiota analysis

Twenty four h after the last gavage, fecal pellets were collected from each mouse and stored in an ultra-low temperature freezer. Microbial DNA was extracted from mouse stool samples. The 16S rRNA gene was amplified using polymerase chain reaction (PCR) thermocycler with primer pairs, and purified and quantified using a Quantus™ Fluorometer.

The original data were cut, filtered and spliced into tags, and the target fragments were further filtered and clustered into operational taxonomic units (OTUs). OTUs with 0.97 similarity cutoff were clustered using UPARSE (version 7.1, http://drive5.com/uparse/), and chimeric sequences were identified and removed. The taxonomy of each OTU representative sequence was analyzed using the RDP Classifier (http://rdp.cme.msu.edu/) against the 16S rRNA database (silva138/16s_bacteria) using confidence threshold of 0.7. The intestinal microbiota diversity analysis was performed on the Majorbio Cloud Platform (https://cloud.majorbio.com/). Common and unique OTUs were statistically analyzed according to the total OTUs obtained, and OTU species annotation was performed to obtain community composition information at different levels for each sample. Based on the community composition information, the alpha diversity of the samples was calculated to determine the richness and uniformity. Through the principal coordinate analysis of the beta diversity, the community structure among different samples and groups were further compared.

3. Statistical analysis

The results are expressed as the mean ± standard deviation. IBM SPSS Statistics 23.0 was used to perform one-way analysis of variance (ANOVA) of the study results, and then Dunnett's *t*-test was performed for multiple comparisons. Statistical significance was set at $P < 0.05$.

4. Results

4.1. LWE fingerprint

The active component of LWE was validated according to the content determination suggested by the Chinese Pharmacopoeia 2020 edition (Chinese Pharmacopoeia Commission, 2020). Two chemical components, liquiritin and glycyrrhizic acid, were selected quality markers for LWE (Fig. 1).

4.2. LWE exhibits protective effects in mice with UC

To assess the effect of LWE in UC mice, the research team evaluated the animal activity and inflammation leading to bleeding. The process of modeling and treating acute colitis is illustrated in Fig. 2A. The observation and comparison showed that the mice without 3% DSS intervention had shiny, lively hair, sensitive movements, and normal food and water intake. However, in the 3% DSS treatment group, the mice had dull hair, lazy activity, lethargy, and significantly reduced food intake with weight loss. Compared with the above groups of mice, the opposite was observed in mice in the LWE intervention groups, which was closely resembled the normal mice.

During the experimental period, the average weight increase of normal mice was significantly, whereas the wight of mice with 3% DSS decreased considerably. All mice in the 3% DSS treatment group displayed bloody stool, which may have been caused by severe colitis. After the LWE intervention, mice in the model group gained weight until day 14 (Fig. 2B).

After DSS-induced inflammation, the length of the colons was significantly shortened compared to that in normal mice (Fig. 3). In contrast, the shortening of the colon length was evidently improved in the LWE high-dose-treated group. There were no remarkable differences between 2 g/kg/d or 4 g/kg/d LWE groups and the model group.

4.3. LWE improves the histopathology of colon tissues and repairs the intestinal mucosa

The results of the HE staining are shown in Fig. 4. The intestinal mucosa of normal mice was intact, and crypt structure and abundant goblet cells were observed, without inflammatory infiltration (Fig. 4A). After treatment with 3% DSS, the colonic tissues displayed marked mucosal ulceration, crypt destruction and inflammatory cell infiltration (Fig. 4B). Nonetheless, LWE intervention significantly improved the lesions in the colonic tissue (Fig. 4D–F).

Tight junction (TJ) proteins were measured using IHC. These proteins were arranged in a compact and orderly manner in the normal mice. The distribution of occludin, claudin-1, and ZO-1 was dispersed and hollow in the model group, indicating that UC induced a decreases in their expression. LWE significantly increased the expression of TJ proteins expression (Fig. 5).

Further observation using TEM revealed the location of TJs as a linear dense thin layer, and the microvilli of the colonic epithelium in normal mice were arranged neatly (Fig. 6A). However, the intestinal morphology in the model group showed loss of microvilli on the surface of the colon (Fig. 6B). Compared to the model group, the length of the intestinal villi in the LWE groups increased significantly, and the distribution was more uniform.

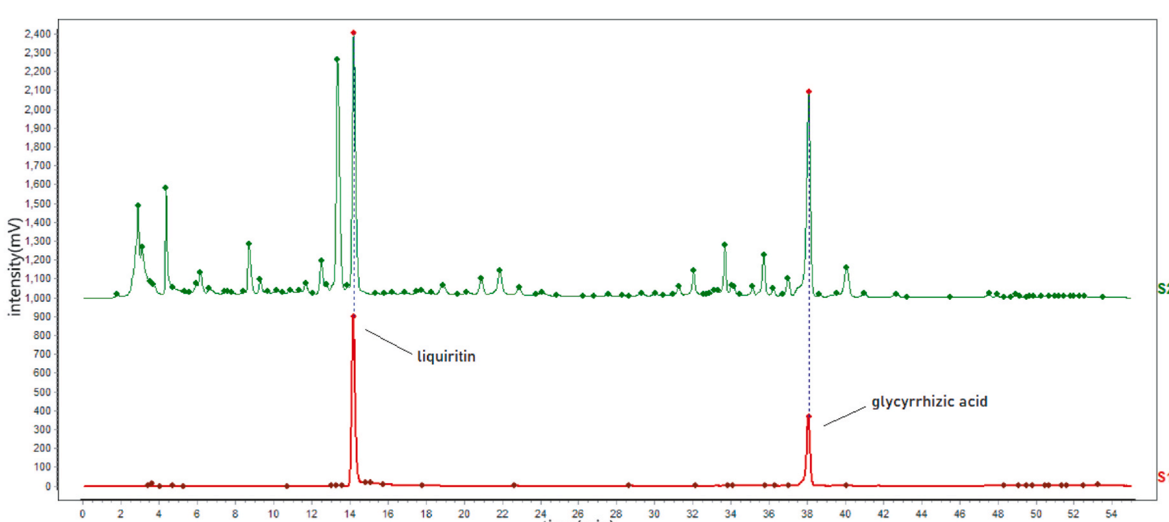

Fig. 1. LWE fingerprint.
HPLC was used to analyze the main components in the LWE. (S1) Fingerprint of LWE. (S2) Mixed standard solution (liquiritin and glycyrrhizic acid).

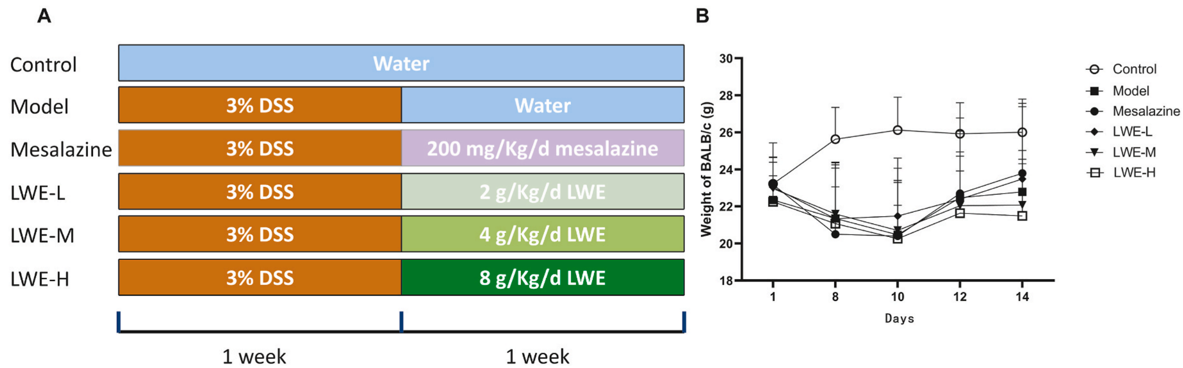

Fig. 2. LWE tends to attenuates DSS-induced weight reduction.
After 7 days of 3% DSS for UC modeling, LWE and mesalazine was administered by gavage for another 7 days. (A) Experimental protocol. (B) Body weight changes. (n = 10)
Control: Normal control; Model: 3% DSS; Mesalazine: 200 mg/kg/d mesalazine; LWE-L: 2 g/kg/d LWE; LWE-M: 4 g/kg/d LWE; and LWE-H: 8 g/kg/d LWE.

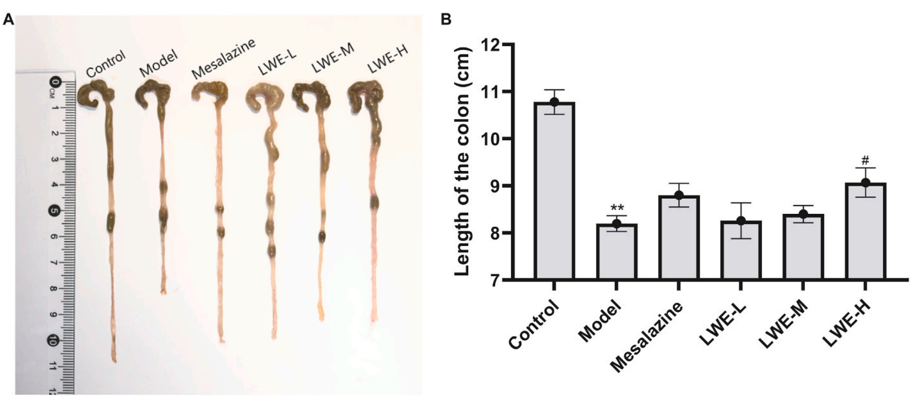

Fig. 3. Colon length. (n = 10)
After 7 days of drug intervention, the colons in each group were obtained and the length was measured. (A) Image of colon tissue. (B) Statistical histogram of colon length. (Control: Normal control; Model: 3% DSS; Mesalazine: 200 mg/kg/d mesalazine; LWE-L: 2 g/kg/d LWE; LWE-M: 4 g/kg/d LWE; and LWE-H: 8 g/kg/d LWE. Compared with control group, $^*P < 0.05$, $^{**}P < 0.01$; compared with model group, $^{\#}P < 0.05$).

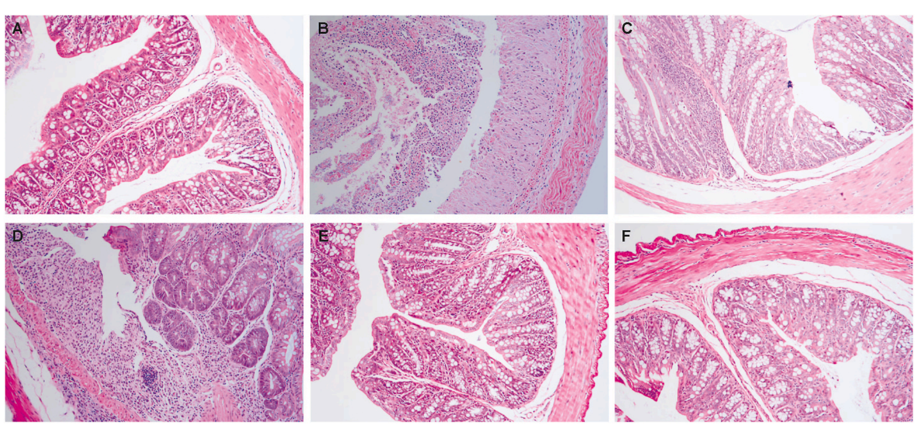

Fig. 4. Histopathological evaluation.
The colons were stained with HE and the pathological changes were observed (magnification = 200 ×). (A) Control group; (B). Model group; (C) 200 mg/kg/d mesalazine; (D) 2 g/kg/d LWE; (E) 4 g/kg/d LWE; (F) 8 g/kg/d LWE.

4.4. LWE regulates inflammatory cytokines in the serum

Next, cytokines content was explored. Levels of IL-1β, IL-8, IL-18, TNF-α, and IL-10 were detected using ELISA kits. Fig. 7 shows that the cytokines levels of IL-1β, IL-8, IL-18, and TNF-α in the 3% DSS-induced mice was raised evidently than those in the normal mice, the cytokines levels of IL-10 was significantly decreased. After 7 d of LWE exposure, these cytokines content in each LWE dose group was obviously lower than that without LWE intervention, meanwhile, LWE high dose group raised the level of IL-10.

4.5. LWE reverses the imbalance of Th17/Treg cell differentiation

The ratio of T helper cells in the colon was determined using flow cytometer. As shown in Table 1 and Fig. 8A, Th17 cells were dramatically expanded during colitis, while Treg cells were reduced in the 3% DSS-induced mice. Fig. 8B shows that the Th17/Treg ratios of UC mice were clearly higher than those of normal mice. Notably, the Th17/Treg cell ratio was significantly suppressed in the LWE group. Further analysis of the expression of ROR-γt and Foxp3 proteins showed that the results were consistent with the flow cytometry data (Fig. 8C and D).

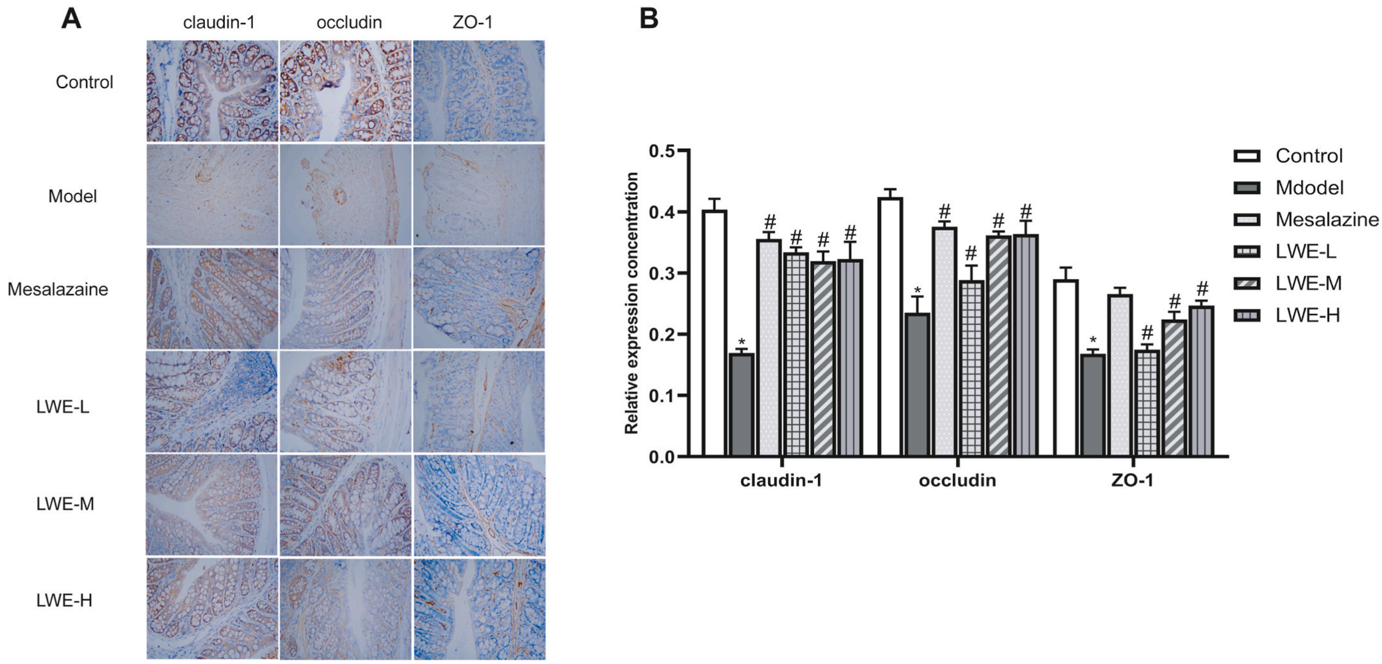

Fig. 5. TJ proteins were measured using IHC.
(A) IHC images of tissues (magnification = 400 ×). (B) Relative expression levels of occludin, claudin-1 and ZO-1 proteins. (Control: Normal control; Model: 3% DSS; Mesalazine: 200 mg/kg/d mesalazine; LWE-L: 2 g/kg/d LWE; LWE-M: 4 g/kg/d LWE; and LWE-H: 8 g/kg/d LWE).

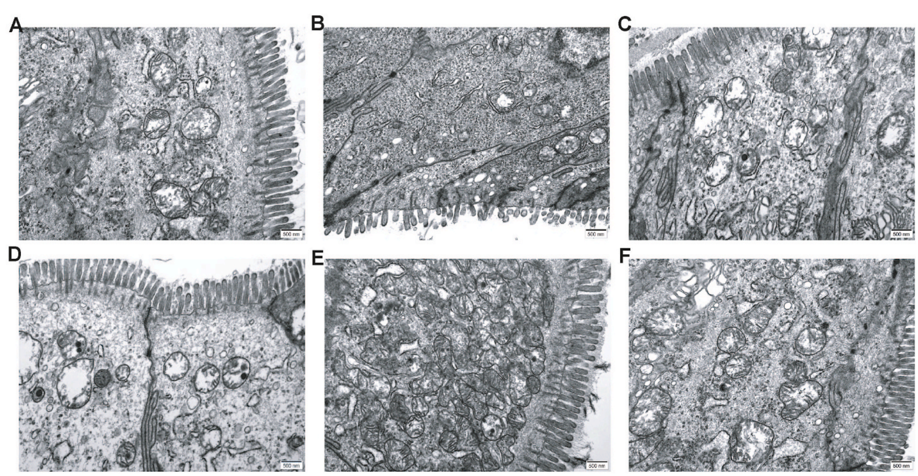

Fig. 6. Ultrastructural changes of microvilli on the surface of the colon using TEM. (magnification = 25, 000 × , bar = 500 nm).
(A) Control group; (B). Model group; (C) 200 mg/kg/d mesalazine; (D) 2 g/kg/d LWE; (E) 4 g/kg/d LWE; (F) 8 g/kg/d LWE.

4.6. Modulating effect of LWE on the intestinal microbiota of UC mice

To fully understand the distribution of OTUs among each group, we constructed Venn diagrams and used them to calculate the proportion of unique and shared OTUs between the different groups (Fig. 9A).

Beta-diversity was used to measure the differences in the community composition of the gut microbiota. At the genus level, PCoA and PLS-DA demonstrated close distances, implying similar species compositions between the two groups (Fig. 9B and C).

Phylum level analysis revealed *Firmicutes* and *Bacteroidetes* were the dominant bacteria (Fig. 10A and B). The gut communities of mice in the control group were dominated by *Bacteroidetes* (55%, mean abundance) and *Firmicutes* (36.03%), whereas the model group samples had relative abundances of *Bacteroidetes* (28.74%) and *Firmicutes* (62.35%). Interestingly, the LWE groups had significant abundances of *Verrucomicrobiota* (3.73%~12.99%). The abundance of *Bacteroidetes* decreased with increasing LWE dose, the abundance of *Verrucomicrobiota* increased with increasing LWE dose, and the LWE high-dose exhibited low abundance. At the genus level, linear discriminant analysis coupled with effect size measurements (LEfSe) was performed to determine the differentially abundant microbial genera in each group of UC mice. As shown in the results, g_*Lachnospiraceae*_NK4A136_group and c_*Bacilli* were significantly enriched in the normal mice and UC mice, respectively. g_*Akkermansia* and p_*Verrucomicrobiota* played vital roles in the LWE medium-dose group (Fig. 10C and D). In addition, bacterial species among different groups were compared at the species level, and the top five species with significant differences in content between the control and model groups were *Muribaculaceae*, *Lachnospiraceae*_NK4A136_group, *Alistipes*, *Candidatus_Saccharimonas*, and *Rikenellaceae*_RC9_gut_group (Fig. 10E). Simultaneously, the relative abundances of *Akkermansia muciniphila*, norank_f_*Muribaculaceae*, *Candidatus Saccharimonas*, and *Enterorhabdu* were dramatically higher in the

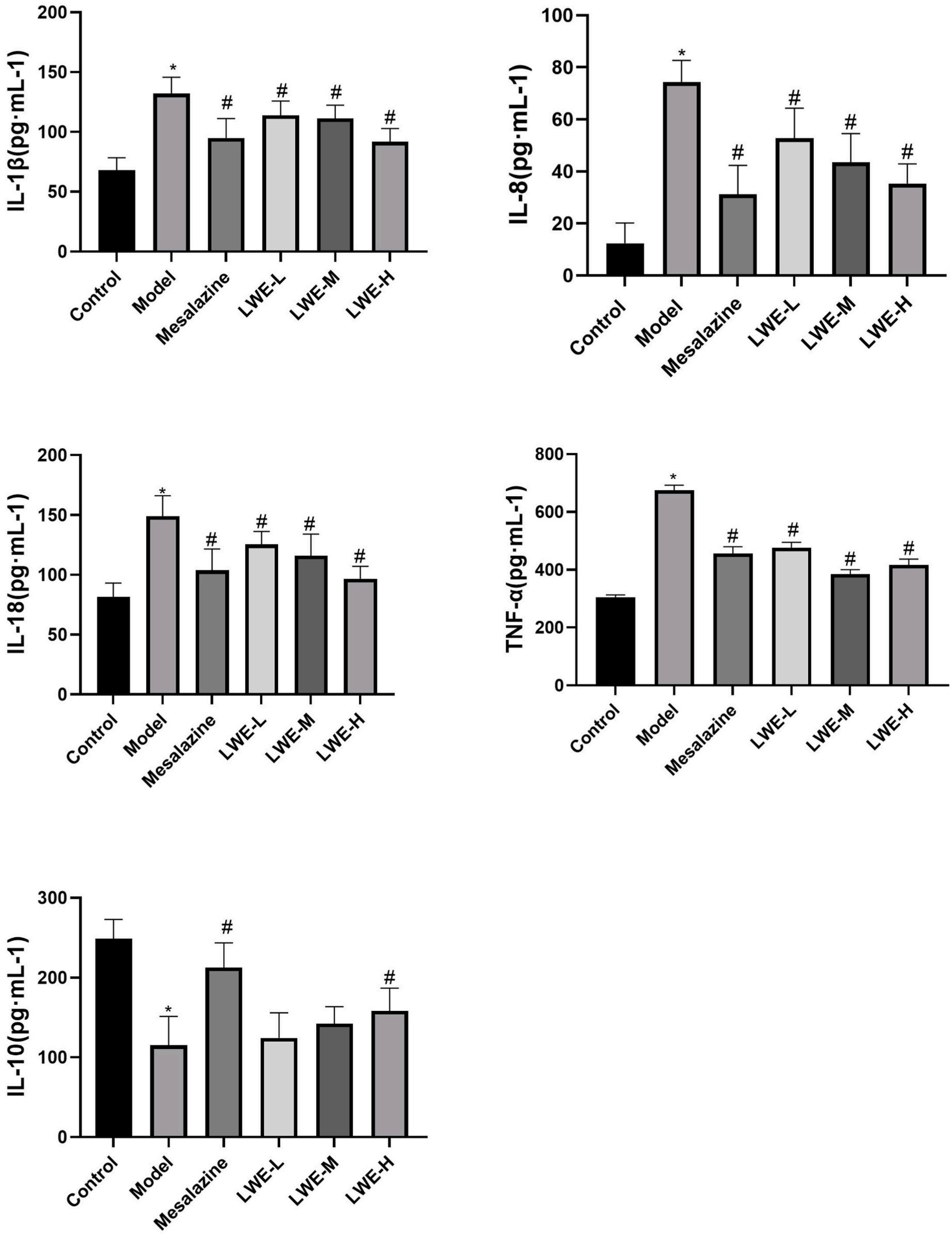

Fig. 7. Effects of LWE on IL-1β, IL-8, IL-18, TNF-α and IL-10 in the colonic tissues. (n = 3)
(Control: Normal control; Model: 3% DSS; Mesalazine: 200 mg/kg/d mesalazine; LWE-L: 2 g/kg/d LWE; LWE-M: 4 g/kg/d LWE; and LWE-H: 8 g/kg/d LWE. Compared to the control group, $^{*}P < 0.05$; compared to the model group, $^{\#}P < 0.05$).

LWE medium-dose group (4 g/kg/d) than in the model group (Fig. 10F).

According to the abundance variation in the gut microbiota, the functional changes were predicted using PICRUSt. As shown in Fig. 10, the clusters of orthologous groups (COG) and Kyoto Encyclopedia of Genes and Genomes (KEGG) functions of the OTUs were annotated. Changes in the intestinal flora in different groups were closely associated with physiological functions (Fig. 11A). The main functions are metabolism and genetic information processing (Fig. 11B).

4.7. *LWE inhibits the NF-κB signaling cascade*

The expression of NF-κB signaling pathway-related proteins in

Table 1
Analysis of Th17 cells and Treg cells in colon (mean ± SD, n = 3).

Groups	Th17/%	Treg/%
Control group	0.89 ± 0.07	7.34 ± 0.92
Model group	12.43 ± 1.24**	0.87 ± 0.08**
Mesalazine group	6.18 ± 0.84##	4.84 ± 0.28##
2 g/kg/d LWE	9.46 ± 1.58##	2.95 ± 0.28##
4 g/kg/d LWE	10.06 ± 0.78##	2.40 ± 0.38##
8 g/kg/d LWE	8.28 ± 0.70##	4.33 ± 0.65##

Compared with control group, **$P < 0.01$; compared with model group, ##$P < 0.01$.

colonic tissue was significantly increased after DSS exposure; however, LWE reduced the expression of TLR4, MyD88, and NF-κB p65, especially in the 4 g/kg/d LWE group. The results showed that LWE could improved inflammation in DSS-induced UC through inhibition of the TRL4/MyD88/NF-κB signaling pathway (Fig. 12).

5. . Discussion

UC is a common non-specific inflammatory disease. Typical symptoms of experimental UC include weight loss, fecal bleeding and colonic tissue histological damage (Feuerstein et al., 2019). The treatment of patients with UC frequently involves 5-aminosalicylic acid and corticosteroids, which may have serious adverse effects (Nakase et al., 2021). Currently, treatment with TCM is often preferred by UC patients (Chen

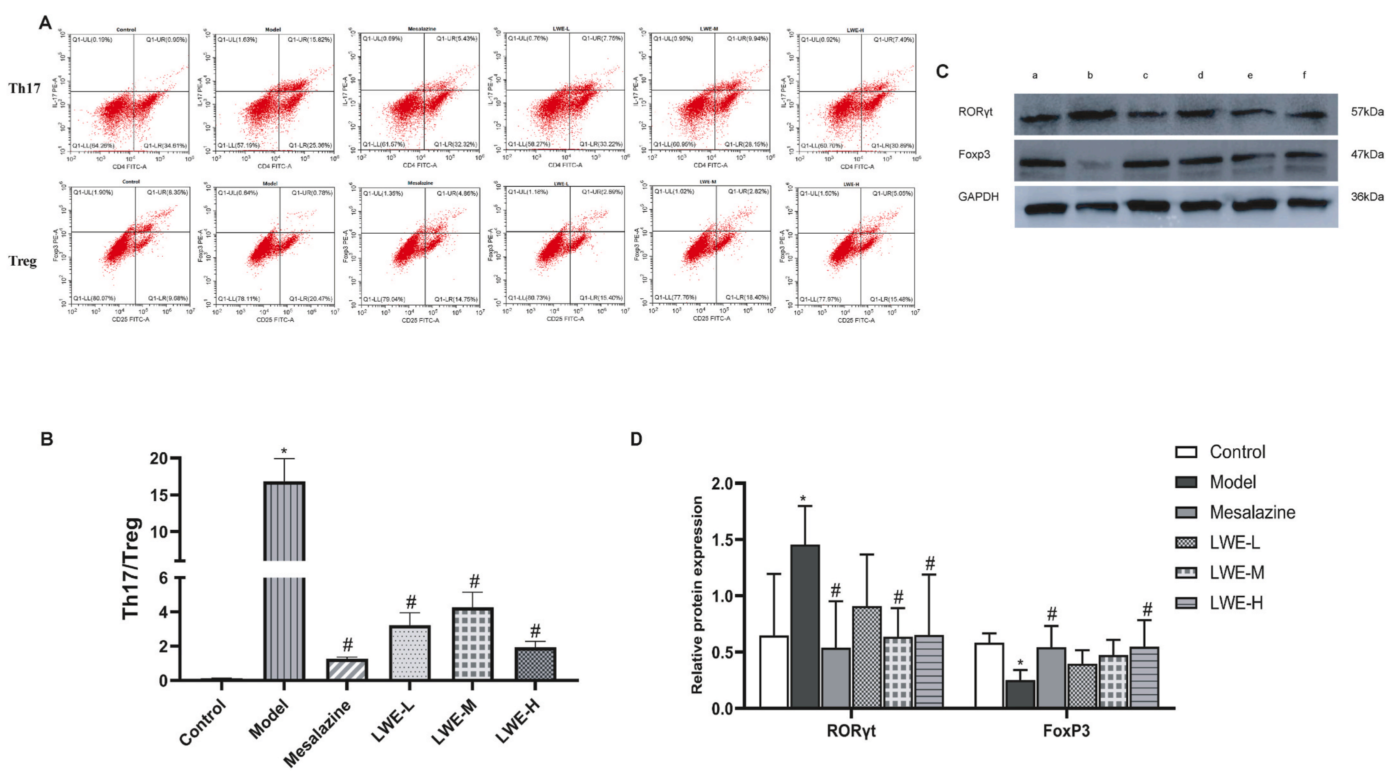

Fig. 8. Th17 cells and Treg cells differentiation.
(A) Quadrant data. (B) Ratio of Treg/Th17 cells in colon. (C) The expression of ROR-γt and Foxp3 were detected using WB (a: Control group; b: Model group; c: 200 mg/kg/d Mesalazine; d: 2 g/kg/d LWE; e: 4 g/kg/d LWE; f: 8 g/kg/d LWE). (D) Relative proteins expression. (Compared to the control group, *$P < 0.05$, **$P < 0.01$; compared to the model group, #$P < 0.05$, ##$P < 0.01$).

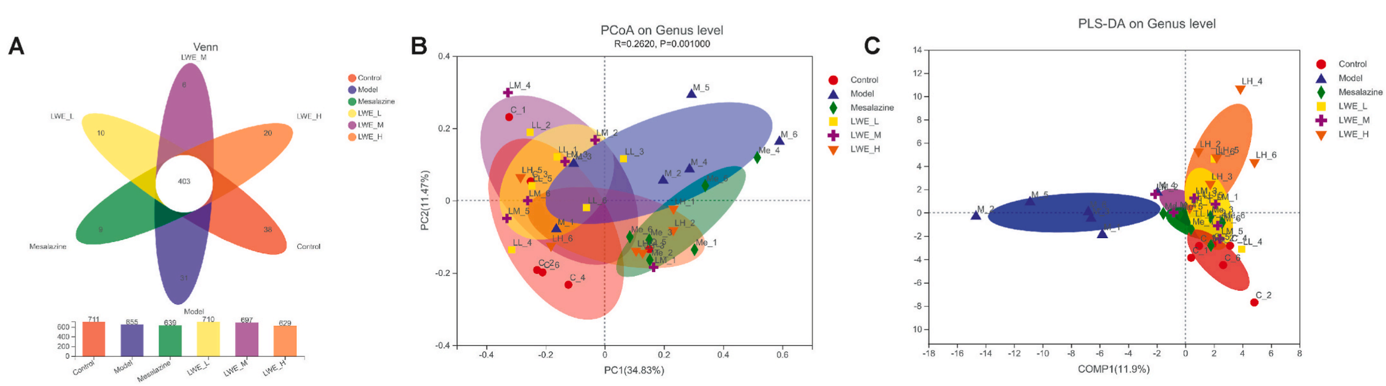

Fig. 9. Venn and Beta diversity measures.
(A) Venn diagrams based on OTUs. (B) PCoA at the genus level. (C) PLS-DA at the genus level. (Control: Nomal control; Model: 3% DSS; Mesalazine: 200 mg/kg/d mesalazine; LWE-L: 2 g/kg/d LWE; LWE-M: 4 g/kg/d LWE; and LWE-H: 8 g/kg/d LWE).

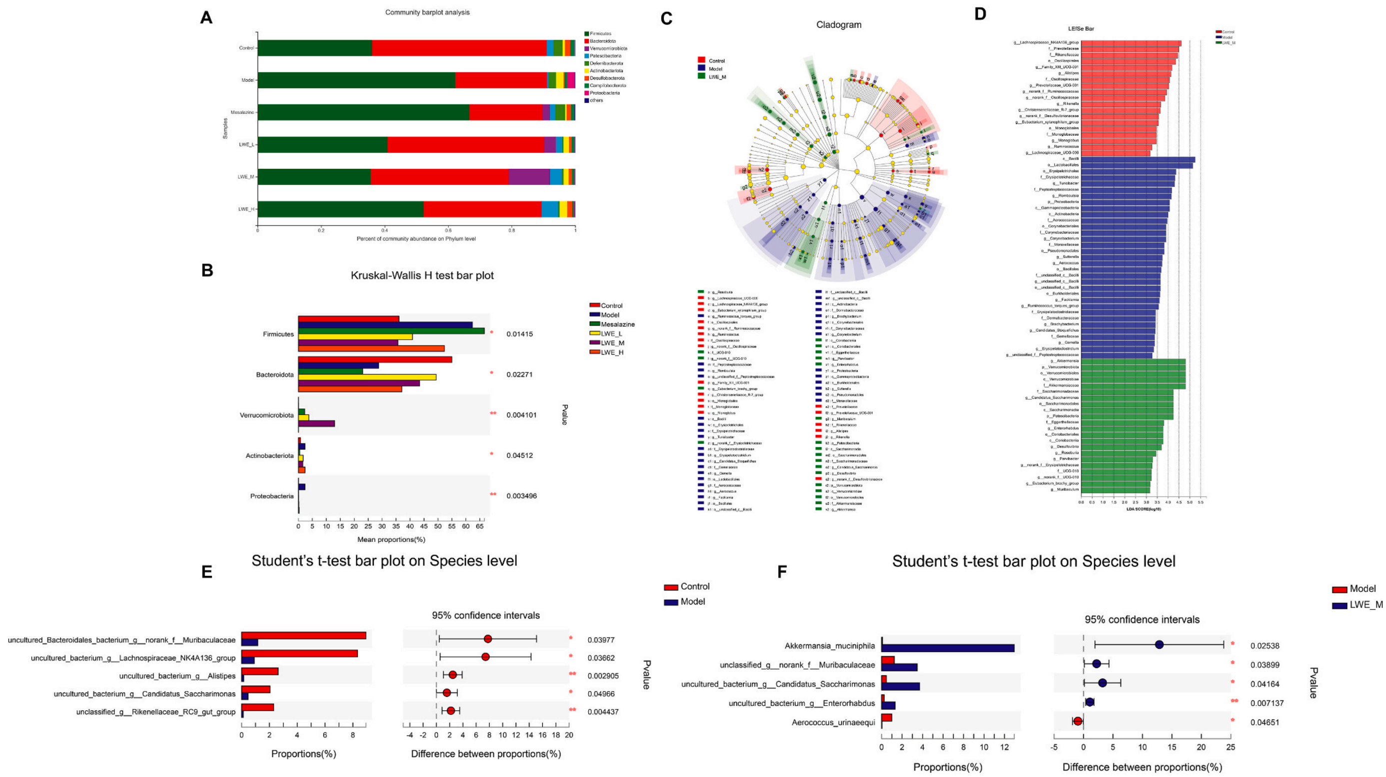

Fig. 10. Community barplot analysis based on phylum level.
(A) Community barplot analysis. (B) Kruskal-Waliis H test bar plot. (C, D) Bacterial taxa from the phylum to genus differentially enriched in mice determined using the LEfSe algorithm. (E) Student's t-test bar plot at the Genus level comparing thecontrol and model groups. (F) Student's t-test bar plot at the Genus level between the model and 4 g/kg/d LWE group. (* 0.01 $<P\leq 0.05$, ** 0.001 $<P\leq 0.01$).

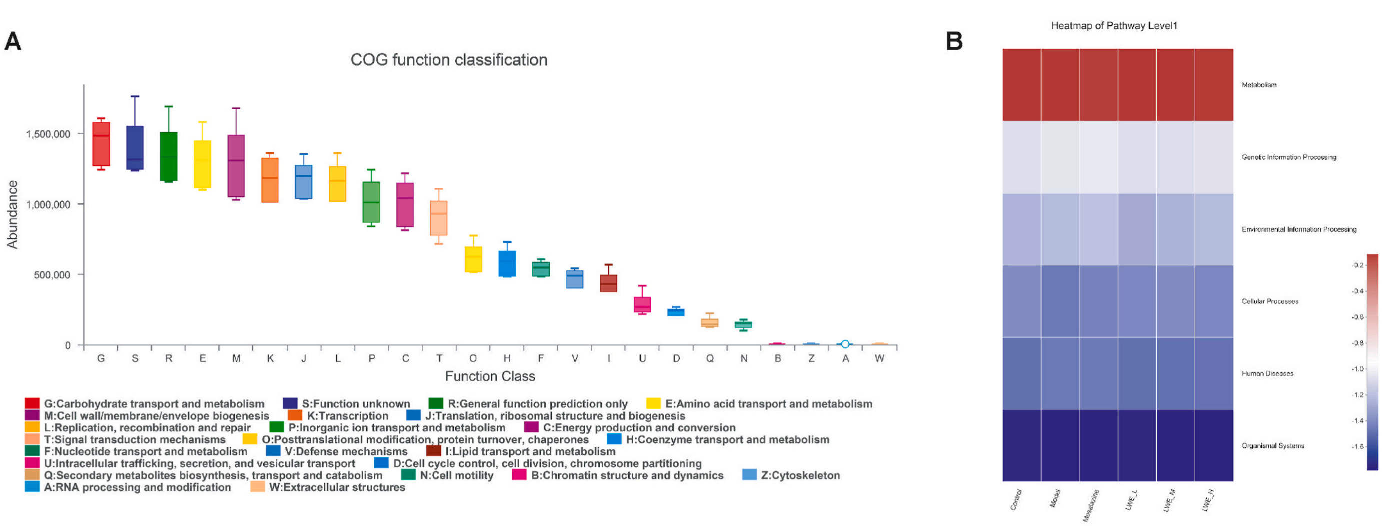

Fig. 11. PICRUSt-predicted microbial community function changes.
(A) COG function classification. (B) Heatmap of pathway level 1.

et al., 2020b; Liu et al., 2022; Xiong et al., 2021).

In this study, we successfully replicated the experimental acute UC mouse model induced by the DSS-free drinking method and treated the mice with LWE by gavage. Results showed that LWE attenuates inflammatory cytokines in the colonic mucosa of 3% DSS-induced UC mice, alleviate pathological damage and hematochezia of the colon in UC mice, significantly increase body weight and colon length, and improves the overall physiological state.

A recent study indicated that the intestinal epithelium plays a central role in IBD, and may respond to inflammatory signals and regulate the protective barrier function of intestine and the composition of flora (Martini et al., 2017). Glycyrrhetinic acid has been reported to promote intestinal epithelial homeostasis through modulation of human antigen R (Chen et al., 2019). Consequently, the study also investigated the pathology of the colonic tissue. The results of the experiments showed that under normal conditions, the intestinal mucosal epithelium of mice was continuous and intact, whereas the model group indicated that the intestinal mucosal epithelium was incomplete, massive inflammatory cell aggregation and infiltration, and the mechanical barrier was damaged. However, it is noteworthy that inflammatory cells in mouse

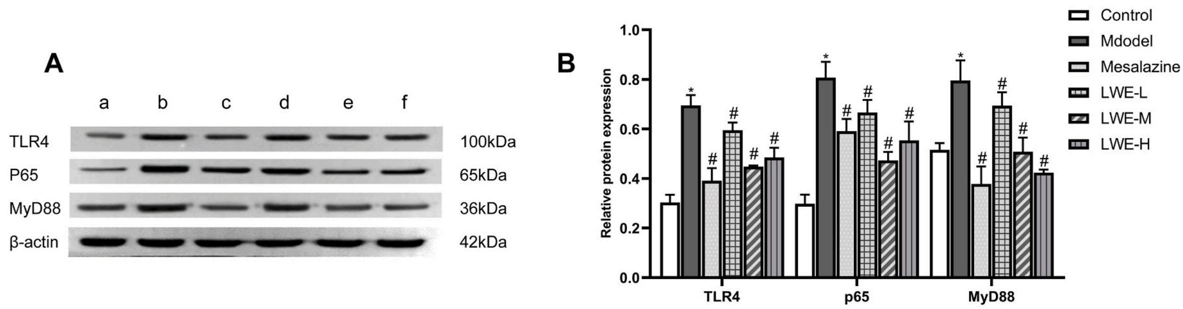

Fig. 12. The expression of NF-κB related proteins.
(A) The expression of TLR4, MyD88, and NF-κB p65(a: Control group; b: Model group; c: 200 mg/kg/d mesalazine; d: 2 g/kg/d LWE; e: 4 g/kg/d LWE; f: 8 g/kg/d LWE). (B) Relative proteins expression. (Compared with control group, $^{*}P < 0.05$; compared to the model group, $^{\#}P < 0.05$).

colon tissues decreased significantly and intestinal mucosal epithelial cells was improved with LWE treatment, indicating that LWE improves the colonic epithelium. Evidence has confirmed that intestinal mucosal healing in UC patients is associated with TJ proteins expression (Tan et al., 2019). An extract of *Glycyrrhiza glabra* protected intestinal barrier integrity by regulating TJ proteins and decreasing TNF-α and MPO levels (Murugan et al., 2022). In addition, IHC staining images showed that the TJs consisting of occludin, claudin-1 and ZO-1, were significantly impaired in the model group. However, LWE reversed the trend of damage trend, which was consistent with the HE staining results. As expected, TEM images more clearly showed that LWE repaired intestinal villi and remediated tissue damage caused by DSS.

Cytokines, as critical lynchpins, cause inflammation in UC because they are potent immunomodulatory proteins with two-way regulatory effects on inflammation in the immune systems and play a vital role in controlling many components of the inflammatory response (Ranson et al., 2017; Verstockt et al., 2018), for instance interleukins and TNF-α, which could results in inflammation and tissue damage (Nakase et al., 2022). IL-8 and IL-18 are common pro-inflammatory cytokines that play crucial roles in inducing the expression of adhesion molecules and enhancing cells cross-talking (ElMahdy et al., 2021; Uchiyama et al., 2022). IL-1β and TNF-α can cause the release of inflammatory proteins and inflammatory mediators, causing a series of intestinal inflammatory reactions and local tissue destruction (Dinallo et al., 2019; Pugliese et al., 2017). The imbalance between pro-inflammatory and anti-inflammatory cytokines that leads to disease perpetuation and tissue destruction, and anti-inflammatory cytokines (IL-10) play significant roles in downregulation of disease progression. The data of this study confirmed that the levels of pro-inflammatory cytokines in serum were significantly reduced by LWE, on the contrary, and levels of IL-10 were significantly increased by high dose LWE, suggesting that LWE has excellent immunomodulatory effects.

NF-κB is a regulator that plays a key role in the process of inflammatory and immune system (Barnabei et al., 2021; Zhang et al., 2021). Additionally, to inquire into the potential mechanism of LWE in the treatment of DSS induced UC, this study evaluated the effect of LWE on protein expression were evaluated. Combined with these results, the cytokines levels, histopathology and intestinal mucosal ultrastructure in the LWE groups significantly improved compared to those in the other groups, indicating that LWE showed a protective therapeutic role against UC through inhibition of NF-κB signaling cascade.

Th17 cells are closely associated with the pathology of inflammation. Treg cells function in immunosuppression through direct contact between cells to inhibit or secrete inhibitory cytokines (Lee, 2018). An imbalance between anti-inflammatory (Treg) and pro-inflammatory (Th17) cells is an important feature of in the pathogenesis of UC (Yan et al., 2020). The roles of these two cells are almost opposite, and once the balance between them is disrupted, they become key factors in causing immune diseases (Yu et al., 2021). To further investigate the effect of LWE on the immune system, Th cells differentiation was assessed. In this study, the proportion of Th17 cells and Treg cells in UC mice increased and decreased respectively. In contrast, the proportion of Th17 cells decreased, and the proportion of Treg cells increased after LWE intervention. ROR-γt and Foxp3 are transcription factors that affect the Th cells differentiation. Meanwhile, the results of WB protein detection were consistent with those of flow cytometry, indicating that LWE can reverse the imbalance in Th17/Treg cell differentiation in colonic tissue.

Recent advances in research imply that the intestine is a crucial component of the human digestive system and important in the immune reactions (Becattini et al., 2021). Gut flora constituents are known to influence intestinal immune responses because the disease can be transferred to naive hosts through fecal transplantation in some mouse models of UC (Chen et al., 2021). Research has shown that a single Chinese herbal medicine and numerous well-known TCM formulas can exert anti-inflammatory effects and effectively improve the intestinal flora in UC mice (Luo et al., 2019; Zhu et al., 2020). To investigate the potential mechanism, the gut microbiota of mice was evaluated. It was observed from the diversity results that the medium-dose group of LWE helped to recover the diversity of the intestinal flora, whereas the high- and low-dose groups reduced the diversity of the flora to varying degrees. In terms of community structure, it can be shown that medium-dose LWE recovered the abundance of *Muribaculaceae*. Moreover, the abundance of several probiotics increased significantly, such as *Akkermansia_muciniphila*, *Muribaculaceae*, norank_f_*Candidatus_Saccharimonas*, and *Enterorhabdu*. *A. muciniphila* has been proven to plays an essential role in the progression of UC (Qu et al., 2021). Ginger corrected the disorder of lower abundance of *Muribaculaceae* in UC mice (Guo et al., 2021). The relative abundance of beneficial bacteria *Lactobacillus* and *Candidatus Saccharimonas* was significantly increased by egg white peptides (Cruz et al., 2020). The results of flora diversity indicate that LWE may exert its protective effect on UC through restoring intestinal flora.

Currently, pharmacotherapies for UC have several many side effects and are expensive. This research suggests that licorice may be a promising alternative to UC treatment to reduce inflammatory reactions and improve immune function based on the regulation of intestinal flora. Nonetheless, the mechanisms of the changes in microbiota structure and how intestinal flora or its metabolites perform the damage repair function remain obscure and need to be further verified through researches.

6. .Conclusion

Licorice is considered to possess anti-ulcer activity and immunomodulatory effects, however its mechanism of action is still not clearly understood. In order to advance toward clinical practice, the water extract of licorice was used to analyze its effect on UC mice. This study highlights that licorice may have a significant anti-inflammatory effect and modulate the differentiation of Th17/Treg cells. Furthermore, the study showed that licorice may improve the composition of intestinal flora and increase the content of beneficial bacteria. These findings have demonstrated that licorice is a potential herbal medicine for UC treatment. Licorice may regulate the expression of the TLR4/MyD88/NF-κB signaling pathway by affecting the diversity of intestinal flora, including the species and number of dominant bacteria. The interaction among licorice, intestinal flora and immunity serves as an experimental basis for the research on the anti-inflammatory and immunomodulatory mechanisms of licorice, and also provides a novel strategy for licorice to coordinate the drug actions of prescriptions.

CRediT authorship contribution statement

Gaoxiang Shi: Investigation, Methodology, Writing – original draft, Writing – review & editing. **Jinrong Kong:** Investigation, Methodology, Data curation. **Yunlai Wang:** Methodology, Supervision**Zihua Xuan:** Methodology, Supervision. **Fan Xu:** Conceptualization, Funding acquisition, Project administration, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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